Lab 5-Arithmetic, Comparison, and Logical Operators

**Creating Database and Doctors Table**

```
CREATE DATABASE IF NOT EXISTS hospital_db;
USE hospital_db;

DROP TABLE IF EXISTS Appointments;
DROP TABLE IF EXISTS Doctors;

CREATE TABLE Doctors (
    doctor_id INT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(50),
    base_salary DECIMAL(10, 2),
    bonus DECIMAL(10, 2),
    allowance DECIMAL(10, 2),
    consultation_fee DECIMAL(10, 2),
    shift_hours INT
);

INSERT INTO Doctors (doctor_id, name, department, base_salary, bonus, allowance,
consultation_fee, shift_hours) VALUES
(101, 'Sahasra', 'Cardiology', 90000.00, 5000.00, 2000.00, 500.00, 8),
(102, 'Ananya', 'Neurology', 110000.00, 8000.00, 3000.00, 600.00, 10),
(103, 'Ninada', 'Pediatrics', 80000.00, 4000.00, 1500.00, 400.00, 6);
```

Output:

```
+-----------+---------+------------+-------------+---------+-----------+------
| doctor_id | name    | department | base_salary | bonus   | allowance | consu
+-----------+---------+------------+-------------+---------+-----------+------
|       101 | Sahasra | Cardiology |    90000.00 | 5000.00 |   2000.00 |
|       102 | Ananya  | Neurology  |   110000.00 | 8000.00 |   3000.00 |
|       103 | Ninada  | Pediatrics |    80000.00 | 4000.00 |   1500.00 |
+-----------+---------+------------+-------------+---------+-----------+------
```

Arithmetic Operations (Addition & Subtraction)

```
SELECT
    doctor_id,
```

```
    name,
    base_salary,
    bonus,
    allowance,
    (base_salary + bonus + allowance) AS total_compensation,
    ((base_salary + bonus + allowance) - 5000.00) AS net_compensation
FROM Doctors;
```

Output:

Output:

| doctor_id | name | base_salary | bonus | allowance | total_compensation | net_compensation |
|---|---|---|---|---|---|---|
| 101 | Sahasra | 90000.00 | 5000.00 | 2000.00 | 97000.00 | 92000.00 |
| 102 | Ananya | 110000.00 | 8000.00 | 3000.00 | 121000.00 | 116000.00 |
| 103 | Ninada | 80000.00 | 4000.00 | 1500.00 | 85500.00 | 80500.00 |

Arithmetic Operations (Multiplication, Division, Modulo)

```
SELECT
    name,
    shift_hours,
    base_salary,
    (base_salary * 12) AS annual_base_salary,
    (base_salary / 160) AS estimated_hourly_rate,
    (shift_hours % 2) AS shift_hour_remainder
FROM Doctors;
```

Output:

Output:

| name | shift_hours | base_salary | annual_base_salary | estimated_hourly_rate | shift_remainder |
|---|---|---|---|---|---|
| Sahasra | 8 | 90000.00 | 1080000.00 | 562.500000 | 0 |
| Ananya | 10 | 110000.00 | 1320000.00 | 687.500000 | 0 |
| Ninada | 6 | 80000.00 | 960000.00 | 500.000000 | 0 |

Comparison Operators (=, >, <>)

```
SELECT doctor_id, name, department, base_salary
FROM Doctors
WHERE department = 'Cardiology' OR base_salary > 85000.00;

SELECT doctor_id, name, department, consultation_fee
FROM Doctors
WHERE department <> 'Pediatrics' AND consultation_fee < 600.00;
```

Output:

Output:

```
+-----------+---------+------------+-------------+
| doctor_id | name    | department | base_salary |
+-----------+---------+------------+-------------+
|       101 | Sahasra | Cardiology |    90000.00 |
|       102 | Ananya  | Neurology  |   110000.00 |
+-----------+---------+------------+-------------+
+-----------+---------+------------+------------------+
| doctor_id | name    | department | consultation_fee |
+-----------+---------+------------+------------------+
|       101 | Sahasra | Cardiology |           500.00 |
+-----------+---------+------------+------------------+
```

Range and List Comparisons (>=, <=, BETWEEN, IN)

```
SELECT name, base_salary
FROM Doctors
WHERE base_salary >= 80000.00 AND base_salary <= 100000.00;

SELECT name, consultation_fee
FROM Doctors
WHERE consultation_fee BETWEEN 400.00 AND 600.00;

SELECT name, department
FROM Doctors
WHERE name IN ('Sahasra', 'Ninada');
```

Output:

Output:

```
| name    | base_salary |
+---------+-------------+
| Sahasra |    90000.00 |
| Ninada  |    80000.00 |
+---------+-------------+
```

```
+---------+------------------+
| name    | consultation_fee |
+---------+------------------+
| Sahasra |           500.00 |
| Ananya  |           600.00 |
| Ninada  |           400.00 |
+---------+------------------+
```

```
+---------+------------+
| name    | department |
+---------+------------+
| Sahasra | Cardiology |
| Ninada  | Pediatrics |
```

Logical Operators (AND, OR, NOT)

```
SELECT name, department, base_salary
FROM Doctors
WHERE base_salary > 85000.00 AND shift_hours >= 8;
```

```
SELECT name, department, consultation_fee
FROM Doctors
WHERE department = 'Neurology' OR consultation_fee = 400.00;
```

```
SELECT name, department
FROM Doctors
WHERE NOT (department = 'Pediatrics');
```

Output:

Output:

```
| name    | department | base_salary |
+---------+------------+-------------+
| Sahasra | Cardiology |    90000.00 |
| Ananya  | Neurology  |   110000.00 |
+---------+------------+-------------+
+--------+------------+------------------+
| name   | department | consultation_fee |
+--------+------------+------------------+
| Ananya | Neurology  |           600.00 |
| Ninada | Pediatrics |           400.00 |
+--------+------------+------------------+
+---------+------------+
| name    | department |
+---------+------------+
| Sahasra | Cardiology |
| Ananya  | Neurology  |
```

Combined Arithmetic & Logical Expressions

```
SELECT
   name,
   department,
   (base_salary + bonus) AS salary_plus_bonus,
   shift_hours
FROM Doctors
WHERE
   ((base_salary + bonus) > 90000.00)
   AND
   (shift_hours > 7 OR name NOT IN ('Ninada'));
```

Output:

Output:

```
+---------+------------+------------------+-------------+
| name    | department | salary_plus_bonus | shift_hours |
+---------+------------+------------------+-------------+
| Sahasra | Cardiology |          95000.00 |           8 |
| Ananya  | Neurology  |         118000.00 |          10 |
+---------+------------+------------------+-------------+
```